

TRACE-V: a PicoRV32 RISC-V core on SKY130, built by agentic AI and verified with independent open-source tools

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TRACE-V: Traceable RISC-V — AI-built, Checked, Evidenced. The release is published at <https://github.com/raisul1212/trace-v> for non-commercial academic instruction and research.

Release v1.0.1 replaces the 8 ns route of v1.0 with an 8.25 ns route, which has a wider setup margin at the slow corner; the 8 ns release remains available at tag v1.0.

Abstract

- The PicoRV32 RV32I core was taken from RTL to block-level signoff on SkyWater sky130A at 8.25 ns (about 121 MHz). The requirements were set and the decisions were approved by the designer; implementation, verification, debugging and fixes were carried out by agentic AI on a commercial implementation flow. The routed block has 7,845 cells in a 357.42×355.64 μm die.
- Setup slack is +0.180 / +2.388 / +2.957 ns and hold slack is +0.716 / +0.214 / +0.034 ns at ss / tt / ff, after 0.25 ns clock uncertainty. Slew, capacitance and fanout violations are 0 at all three corners.
- DRC (magic and KLayout), antenna and LVS are clean, and both LEC steps are equivalent. The post-layout gate-level run of the golden program passes at 8.25 ns and the 4.125 ns control fails; a re-run in Xcelium with flop timing checks enforced also passes at 8.25 ns with 0 violations.
- Static IR drop is 11 mV on VPWR and 11.6 mV on VGND against an 80 mV budget, and power is 13.0 mW (static, vectorless).
- Six defects that each passed a tool's own report were found and corrected by agentic AI (section 5). Signoff is at nominal RC (section 7).
- The release covers the following checks. Timing was signed off at three corners with 0.25 ns clock uncertainty. DRC was run with two independent decks, LVS was run against the routed netlist, and logic equivalence was checked at both synthesis and routing. Post-layout gate-level simulation was run with flop timing checks, and static IR was analyzed. Every physical, DRV and simulation check was shown to fail on a known-bad input before it was trusted. Every reported figure is traceable to frozen, checksummed evidence. The physical, timing and simulation checks can be reproduced from the public package with open-source tools.

Background: PicoRV32

PicoRV32 is a CPU core by Claire Xenia Wolf that implements the RISC-V RV32IMC instruction set. It can be configured as RV32E, RV32I, RV32IC, RV32IM or RV32IMC [1]. It is size-optimized, at 750–2,000 LUTs and 250–450 MHz on Xilinx 7-series FPGAs, and is intended mainly as an auxiliary processor in FPGA and ASIC designs [1]. Its average CPI is about 4, depending on the instruction mix. It is released under the ISC license [1].

It is a general-purpose CPU. RV32I is RISC-V’s base 32-bit integer instruction set: 40 distinct instructions covering integer arithmetic, logic, shifts, compares, loads, stores, branches, jumps and system [2]. M (multiply/divide) and C (compressed) are optional extensions [2].

This build uses PicoRV32’s default parameters, unchanged [1]:

- RV32I with all 32 registers and the cycle and instruction counters;
- no multiply/divide, no compressed instructions, no interrupts and no co-processor interface;
- a non-barrel shifter;
- traps on misaligned accesses and illegal instructions.

The memory interface works as follows: the core asserts `mem_valid` and holds it until the memory asserts `mem_ready`. An optional look-ahead interface (`mem_la_*`) announces a transfer one cycle early [1].

The process is SkyWater SKY130, an open-source 130 nm PDK; this design uses the `sky130_fd_sc_hd` standard-cell library [3].

1 The design and target

The design is PicoRV32, an RV32I RISC-V CPU core by Claire Xenia Wolf, released under the ISC license. The process is SkyWater sky130A, an open PDK, with `sky130_fd_sc_hd` standard cells. The target is a single clock of 8.25 ns (about 121 MHz) with 0.25 ns of clock uncertainty applied at every corner, and a fanout limit of 10 that is enforced.

PicoRV32 here is a 32-bit RISC-V core (RV32I). The routed block holds 87,237 transistors in 7,845 standard cells. The count comes from the sky130 SPICE model of every placed cell; filler, well-tap and antenna-diode cells add none.

Corner	Temperature	Supply
ss	100 °C	1.60 V
tt	25 °C	1.80 V
ff	−40 °C	1.95 V

The final routed block has the following size and activity figures.

Quantity	Value
Die	$357.42 \times 355.64 \mu\text{m}$
Cells	7,845, of which 176 are clock-tree buffers
Cell area	$71,909 \mu\text{m}^2$
Clock-tree insertion delay	0.86–0.96 ns at ss, average 0.92 ns (Innovus CTS report)
Power	13.0 mW, static and vectorless, 20% input activity at 8.25 ns
Fanout limit	10, enforced

2 Flow and signoff bar

Implementation and timing signoff use a commercial flow. Cadence Genus does synthesis, Innovus does place and route, Tempus produces timing and the SDF, Conformal checks logic equivalence, and Voltus computes IR drop.

Physical signoff uses open checkers, so that anyone can re-verify the layout without the commercial tools: magic DRC, KLayout DRC with the sky130A_mr deck, magic antenna check, LVS with magic and netgen, and gate-level simulation in Icarus Verilog using the post-layout SDF. Icarus Verilog applies SDF path delays but not flop timing checks, so the same simulation is repeated in Cadence Xcelium with those checks enforced (section 3).

The signoff bar was set once by the designer, and the work was carried out against it by agentic AI without renegotiation per run:

- 0.25 ns clock uncertainty at ss, tt and ff, included in both setup and hold.
- DRC clean in two independent decks.
- Max transition, max capacitance and max fanout violations at 0.
- Gate-level simulation and both LEC steps pass.
- Static IR drop within budget, 80 mV on each rail (5% of 1.6 V).

The workload for simulation is a hand-assembled RV32I golden program of 9 instructions that sums 1 to 10 and stores 55.

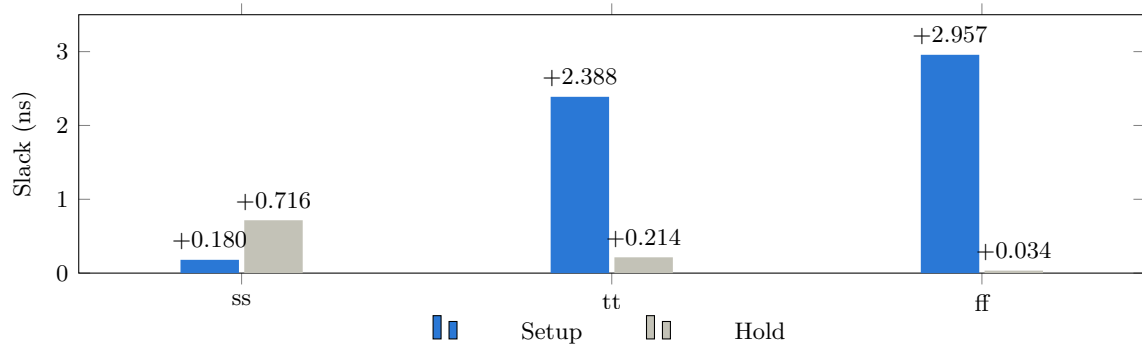
3 Results

Every check below passes at 8.25 ns. Setup and hold both include the 0.25 ns clock uncertainty.

Check	Result
DRC, magic	0 violations
DRC, KLayout	0 findings
Process antenna, magic	0 violations
LVS, magic + netgen, against the routed netlist	Matched; 8,492 / 8,492 nets
LEC, RTL → synthesized netlist (Conformal)	Equivalent; 1,916 compare points, 0 non-equivalent, aborted or unknown
LEC, synthesized → routed netlist (Conformal)	Equivalent; 1,864 compare points, 0 non-equivalent, aborted or unknown
Setup slack, ss / tt / ff	+0.180 / +2.388 / +2.957 ns; 0 violating endpoints
Hold slack, ss / tt / ff	+0.716 / +0.214 / +0.034 ns; 0 of 1,626 endpoints violating, input ports included
Max transition / capacitance / fanout	0 / 0 / 0 at all three corners
Golden program on RTL	Pass
Golden program, post-layout gate level, ss SDF, 8.25 ns	Pass; the 4.125 ns control fails
Gate-level simulation with timing checks (Xcelium), 8.25 ns	Pass, 0 violations; the 4 ns control fails (76 violations)
Post-layout timing wall	Passes at 8 ns, fails at 6 ns
Static IR drop, VPWR / VGND	11 mV / 11.6 mV; budget 80 mV each (5% of 1.6 V)

Setup and hold slack per corner: ss setup +0.180 ns, ff hold +0.034 ns

Slack in ns at 8.25 ns, after 0.25 ns clock uncertainty



Setup and hold slack per corner at 8.25 ns, including 0.25 ns clock uncertainty; values from the signoff results table

Each simulation result has a control: the golden program passes at 8.25 ns and the 4.125 ns control fails, and the post-layout timing wall passes at 8 ns and fails at 6 ns.

Post-layout gate-level simulation was re-run in Cadence Xcelium with flop timing checks enforced, using the same netlist, SDF and testbench. Every timing check in the SDF was applied: 3,114 setup/hold checks and 3,114 pulse-width checks (100%). At 8.25 ns the program passes with 0 timing-check violations, and at 10 ns it passes with 0. At 4 ns, the control, it times out with 76 violations.

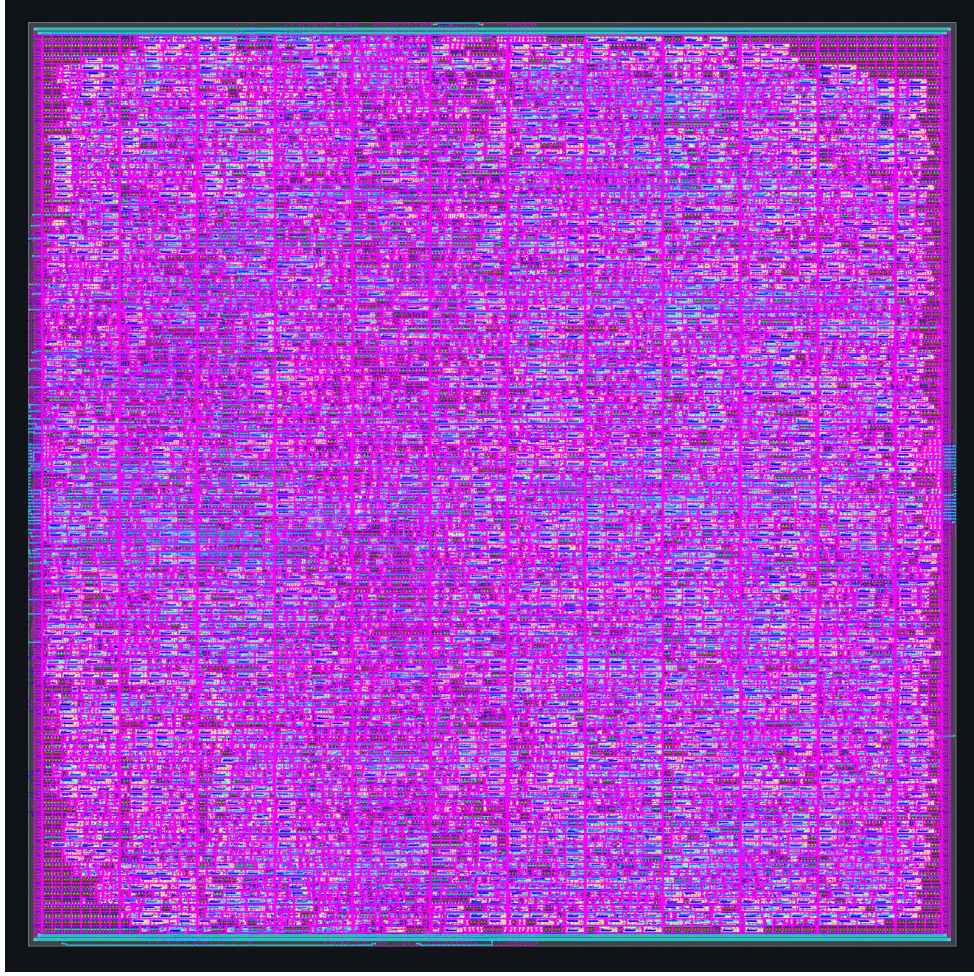


Figure 1. Routed PicoRV32, $357.4 \times 355.6 \mu\text{m}$, sky130A (KLayout render of the final GDS).

4 How the platform works

The requirements were set and the decisions were approved by the designer. Implementation, verification, debugging and fixes were carried out by agentic AI against the signoff bar fixed in section 2. Five properties determine whether the results can be trusted.

1. **Every number traces to its source.** Every figure is traceable to the run and input files that produced it. Nothing is quoted from memory or from a log summary.
2. **Evidence is frozen.** Cited evidence is frozen and checksummed. The final signoff has 75 frozen evidence files (66 at signoff plus 9 from the Xcelium re-run), all verified, with 0 mismatches.
3. **Every physical, DRV and simulation check has a control.** A check is trusted after it has been shown to fail on a known-bad input. Gate-level simulation must fail at 4 ns. The antenna check must fail on a frozen known-bad layout. The slew, capacitance and fanout check must flag a deliberately overloaded net (a 5 pF load). The KLayout deck must fail on a planted violation. A check that passes every input is treated as broken.

4. **Refusal over silence.** A check that cannot run, or whose output cannot be read, is counted as a failure and never as a pass.
5. **Independent checkers.** The tool that builds something is always checked by at least one other tool, because a tool checking its own work shares its own blind spots. Innovus builds the layout, but DRC is run by magic and by KLayout, and LVS by magic and netgen. LVS proves the layout matches the routed netlist; Conformal LEC then proves the routed netlist equals Genus' netlist, and Genus' netlist equals the RTL. The result is an unbroken chain of independent checks from the GDS back to the source code.

5 What the platform caught

Each of the six defects below passed its tool's own report: the flow finished and the tool reported success. For each defect, the symptom, how it was found, the root cause and the fix are given. Three practices exposed the defects:

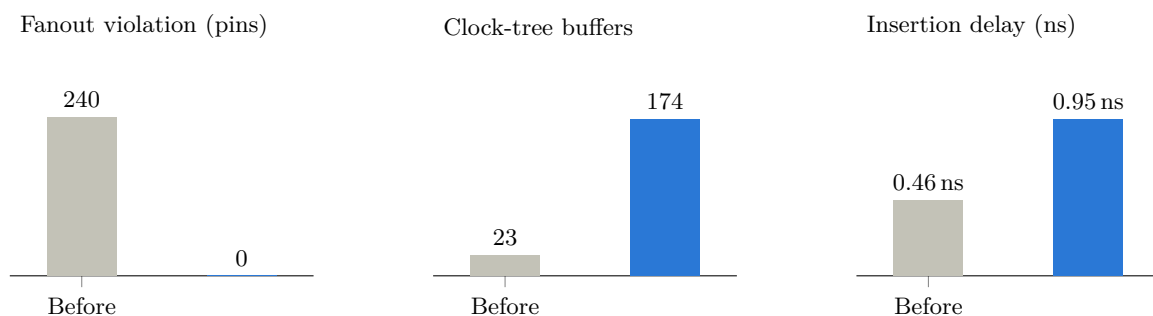
- **Read back.** Each result was read back from the design and compared with the requirement.
- **Cross-checked.** Results that should agree were compared.
- **Controlled.** Each check was shown to fail on a known-bad input before it was trusted.

5.1 The fanout limit was declared and ignored

- **Symptom.** The SDC set a maximum fanout of 10, yet Innovus finished with a 240-pin fanout violation. The tool's flow completed normally.
- **How found.** It was flagged by the slew, capacitance and fanout check. That check is trusted because it flags a deliberately overloaded net (a 5 pF load).
- **Root cause.** The router's optimizer and the clock-tree builder each need fanout repair enabled explicitly.
- **Fix and numbers.** With repair enabled in both, the violation count is 0. The clock tree grew from 23 to 174 buffers, and insertion delay went from 0.46 ns (CTS report average).

Fanout repair took the violation from 240 to 0 and grew the clock tree

Each panel has its own scale; bars start at zero



Fanout-repair fix, before and after; one panel per measure, each on its own scale

5.2 STA passed while the gate-level simulation trapped

- **Symptom.** After the fanout fix, setup was met at 8 ns, but the post-layout simulation hit a trap at 8 ns.
- **How found.** The failing transaction that occurred earliest in the run was traced by agentic AI. The core misread an instruction it had been given correctly. The interface was then timed: the testbench drives memory on the falling edge, so `mem_ready` has half a cycle, and the path `mem_ready` $\rightarrow$ `reg_op1` was -1.87 ns.
- **Root cause.** The constraints set no input or output delays, so STA gave those paths a full cycle. The larger clock tree had made them longer.
- **Fix and numbers.** The SDC now states the environment the testbench models, with half-period input and output delays. The simulation then passes at 8 ns, and the input-port hold violations, worst -0.92 ns before, are gone.

5.3 The gate-level simulation was all-X

- **Symptom.** Every simulation failed, at every clock.
- **How found.** A zero-delay positive control: the same netlist with no delays must pass, and it did not.
- **Root cause.** The power-aware cell models had unconnected supplies, so every output resolved to X.
- **Fix.** The simulated netlist now ties every cell's supply and body pins (VPWR, VGND, VPB, VNB) to constant 1 or 0 before simulation. A zero-delay run of the same netlist must pass; that positive control would have caught the all-X result.

5.4 The antenna check could not fail

- **Symptom.** The initial negative control for the antenna check passed. It contained a diode that was not needed, so it was not a violation, and the check had not been shown able to fail.
- **Root cause.** The control was not a known-bad input.
- **Fix.** It was replaced with a frozen known-bad layout that must fail. The final design has 0 antenna violations.

5.5 A cell excluded from synthesis was re-inserted by place and route

- **Symptom.** KLayout reported one npc.2 violation ($0.16\text{ }\mu\text{m}$ against $0.27\text{ }\mu\text{m}$) on the initial 8.25 ns route, while magic DRC was clean.
- **How found.** The violation was reported by the second, independent DRC deck.
- **Root cause.** The cell `sky130_fd_sc_hd__and2_0` has an NPC edge $0.16\text{ }\mu\text{m}$ inside its boundary. It was excluded in Genus, but restructuring in Innovus re-inserted it.
- **Fix.** The exclusion is applied in Innovus as well, and a post-route count of excluded cells must be zero.

5.6 An antenna violation remained after the router’s repair

- **Symptom.** One net had an antenna ratio of 403.85 against a limit of 400.
- **How found.** The violation was reported by the open antenna check.
- **Root cause.** The router’s repair did not bring the net below the limit.
- **Fix.** The router now repairs to 96% of the limit.

6 Power and IR

Static IR drop from Voltus is 11 mV on VPWR and 11.6 mV on VGND. The budget is 80 mV on each rail, 5% of 1.6 V, so the block uses about 14% of the budget on VPWR. The VPWR map spans 1.589 to 1.600 V, with a worst drop of 11 mV. Power is 13.0 mW, static and vectorless, assuming 20% input activity at 8.25 ns.

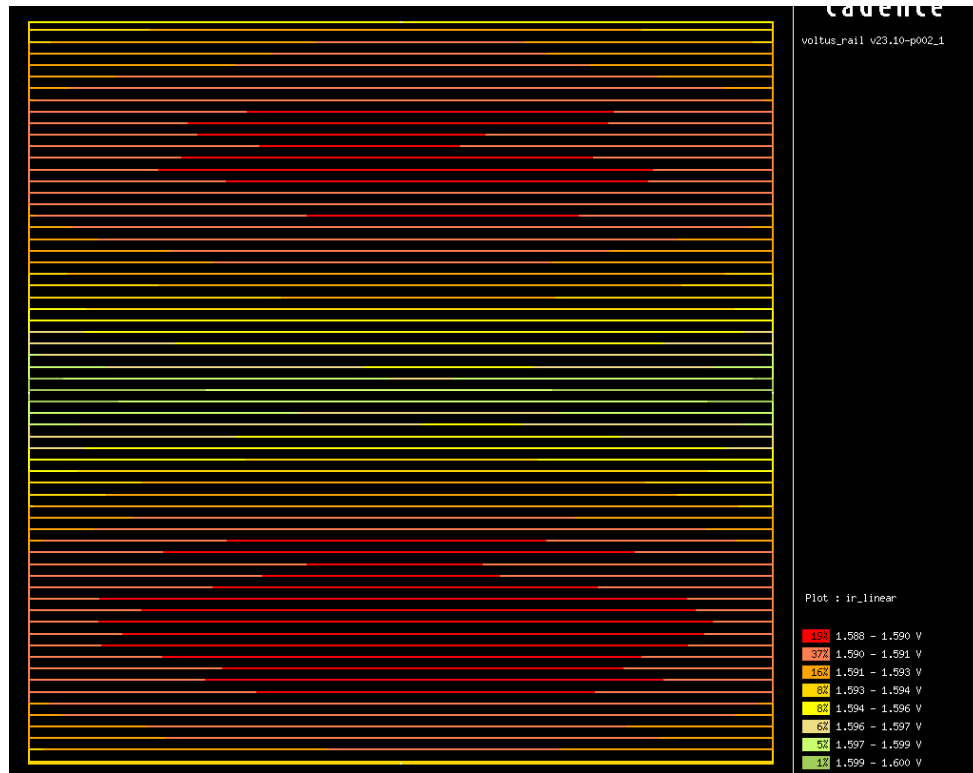


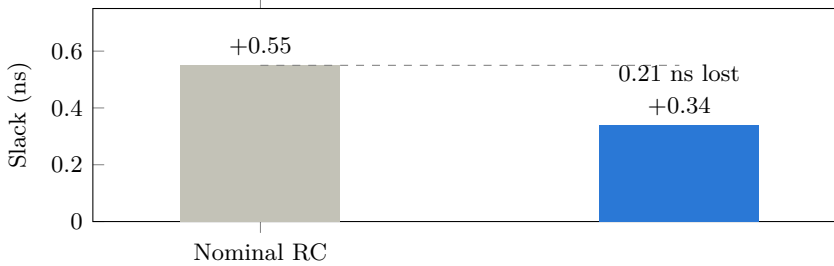
Figure 2. Static IR map of VPWR (Voltus, vectorless): 1.589–1.600 V, worst drop 11 mV against an 80 mV budget.

7 Independent RC cross-check

Signoff extraction is at nominal RC, because the Cadence sky130 kit ships a typical extraction file and no other corner. To bound what that leaves out, the final layout was re-extracted with OpenRCX and timed with OpenSTA, tools independent of the signoff flow.

Max RC costs 0.21 ns of ss setup slack in the OpenRCX and OpenSTA check

ss setup slack at 8.25 ns, in ns; bars start at 0



ss setup slack at 8.25 ns, nominal against max RC; OpenRCX re-extraction timed in OpenSTA

ss setup at 8.25 ns is +0.55 ns with nominal RC and +0.34 ns with max RC, so max RC costs 0.21 ns. The ff hold check with min RC is +0.03 ns and is met.

The figures of about -0.03 ns at max RC and about 8.3 ns to close are an estimate combining two extractors: the 0.21 ns shift measured with OpenRCX and OpenSTA is applied to the signoff margin of +0.180 ns.

8 Cost

All counts run from 1 Oct through the 8 ns signoff (7 Oct, 16:15 EDT), and the figures are held as frozen evidence. The 8.25 ns re-run added 21 cluster jobs and 2.0 h of wall time.

Item	Figure
Calendar time	1 to 7 Oct 2026, from the start of the commercial flow on the open sky130 PDK to block signoff
One full signoff pass, synthesis through the clock sweep	38 min of compute, 45 min end to end
Commercial cluster jobs	62 jobs, 5.1 h total wall time
Agentic AI calls	2,907
Output tokens	1.26 M
Input tokens	0.89 B cached tokens read, 32 M written to cache, about 6.5 k uncached

About 3.1 hours of designer interaction were spent over 7 days, in 86 messages. The requirements were set and each decision was approved by the designer. Implementation, verification, debugging and fixes were carried out by agentic AI. Times were measured from message timestamps, with gaps under 20 minutes counted as work. Token and time figures are upper bounds, because parallel work outside this design was included.

9 Limitations

Packaging, electromigration, metal fill, pad frame, foundry precheck and dynamic IR were outside the scope of this block-level release.

1. **Nominal RC.** Signoff is at nominal RC (section 7).
2. **I/O timing.** I/O timing is set to the testbench’s half-cycle memory protocol. The look-ahead outputs (`mem_la_*`) and the other outputs that the testbench does not sample are unconstrained.
3. **Thin margins.** The margins are ss setup +0.180 ns and ff hold +0.034 ns, after 0.25 ns uncertainty.
4. **Workload.** The post-layout simulation runs one 9-instruction program covering 6 of 40 RV32I opcodes; LEC and STA cover the whole design.

10 Reproducing

The repository at <https://github.com/raisul1212/trace-v> contains:

- rtl and netlists;
- layout: the GDS, byte-identical to the one used at signoff, and the DEF;
- constraints, SPEF and SDF;
- the testbench;
- verify/ scripts for the open-source tools;
- the reports from the open-tool verification run;
- docs/ with the presentation and whitepaper PDFs;
- README, LICENSE, NOTICE and SHA256SUMS.

The physical checks can be re-run with the open-source tools and the scripts in `verify/`. The commercial-tool signoff figures (Genus, Innovus, Tempus, Conformal, Voltus and Xcelium) are stated in this paper and are not regenerated by the open scripts.

License. The released design files are licensed for non-commercial academic instruction and research; commercial use needs written permission. `picorv32.v` keeps its ISC license, and the SKY130 PDK data keep Apache-2.0. For research collaboration or licensing inquiries about the framework, contact Raisul Islam, raisul@purdue.edu.

References

- [1] C. X. Wolf, “PicoRV32 — A Size-Optimized RISC-V CPU,” YosysHQ, <https://github.com/YosysHQ/picorv32> (README).

[2] A. Waterman and K. Asanović (eds.), “The RISC-V Instruction Set Manual, Volume I: Unprivileged ISA,” RISC-V International.

[3] SkyWater Technology and Google, “SkyWater SKY130 PDK,”
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